

Advanced Quantum Mechanics Assignment 3

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Problem 1

This is manifestly a three-degree-of-freedom system, thus, we start by writing the dynamical laws using the Newton's second law of motion:

$$m\partial_t^2 x_1 = -kx_1 + k(x_2 - x_1) \quad (1)$$

$$m\partial_t^2 x_2 = -k(x_2 - x_1) + k(x_3 - x_2) \quad (2)$$

$$m\partial_t^2 x_3 = -k(x_3 - x_2) - k(x_3) \quad (3)$$

Let us write this equations in terms ofn vectors and matrices, and also defining ($\alpha = \frac{k}{m}$).

$$\partial_t^2 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -2\alpha & \alpha & 0 \\ \alpha & -2\alpha & \alpha \\ 0 & \alpha & -2\alpha \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \quad (4)$$

Next, we start to find the basis, in which the matrix above is diagnoalized, to do so we must solve the matrix for its eigenvalues:

$$\det(\Omega - \lambda I) = 0 \quad (5)$$

This would result in the following polynomial equation:

$$-4\alpha^3 - 10\alpha^2\lambda - 6\alpha\lambda^2 - \lambda^3 = 0. \quad (6)$$

Solving this for λ we have:

$$\lambda_1 \rightarrow -2\alpha, \quad \lambda_2 \rightarrow -\alpha(2 - \sqrt{2}), \quad \lambda_3 \rightarrow -\alpha(2 + \sqrt{2}) \quad (7)$$

What this means is that there's a basis in which x'_1, x'_2, x'_3 follow this dynamical law:

$$\begin{cases} \partial_t^2 x'_1 = -2\alpha x'_1 \\ \partial_t^2 x'_2 = -\alpha(2 - \sqrt{2})x'_2 \\ \partial_t^2 x'_3 = -\alpha(2 + \sqrt{2})x'_3 \end{cases} \quad (8)$$

Defining

$$\omega_1 := 2\alpha, \quad \omega_2 := \alpha(2 - \sqrt{2}), \quad \omega_3 := \alpha(2 + \sqrt{2}). \quad (9)$$

Thus we've got three de-coupled ordinary differential equations to solve:

$$\partial_t^2 x'_1 = -\omega_1 x'_1 \quad (10)$$

$$\partial_t^2 x'_2 = -\omega_2 x'_2 \quad (11)$$

$$\partial_t^2 x'_3 = -\omega_3 x'_3, \quad (12)$$

which have the solutions below:

$$x'_1 = A_1 \cos(\sqrt{\omega_1}t + \phi_1) \quad (13)$$

$$x'_2 = A_2 \cos(\sqrt{\omega_2}t + \phi_2) \quad (14)$$

$$x'_3 = A_3 \cos(\sqrt{\omega_3}t + \phi_3) \quad (15)$$

Now it's a good time to notice that x'_1, x'_2 , and x'_3 can be written in terms of our first variables x_1, x_2, x_3 since we know that they satisfy the relation:

$$\Omega x'_i = \lambda_i x'_i \quad (16)$$

And we have the matrix Ω in the original basis, we then have:

$$\begin{pmatrix} -2\alpha & \alpha & 0 \\ \alpha & -2\alpha & \alpha \\ 0 & \alpha & -2\alpha \end{pmatrix} x'_i = \lambda_i x'_i \quad (17)$$

Solving for each (λ_i, x'_i) we find each vector in the x_1, x_2, x_3 basis as:

$$x'_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \quad x'_2 = \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix} \quad x'_3 = \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix} \quad (18)$$

This can be written more nicely in Dirac notation:

$$|1'\rangle = -|1\rangle + |3\rangle \quad (19)$$

$$|2'\rangle = |1\rangle + \sqrt{2}|2\rangle + |3\rangle \quad (20)$$

$$|3'\rangle = |1\rangle - \sqrt{2}|2\rangle + |3\rangle \quad (21)$$

Using this we solve for $|1\rangle, |2\rangle, |3\rangle$, leading to:

$$|1\rangle = \frac{1}{4} (-2|1'\rangle + |3'\rangle + |2'\rangle) \quad (22)$$

$$|2\rangle = \frac{1}{4} (-\sqrt{2}|3'\rangle + \sqrt{2}|2'\rangle) \quad (23)$$

$$|3\rangle = \frac{1}{4} (2|1'\rangle + |3'\rangle + |2'\rangle) \quad (24)$$

Substituting and we have the solutions:

$$|1\rangle = \frac{1}{4} \left(-2A_1 \cos\left(\sqrt{\frac{2k}{m}}t + \phi_1\right) + A_3 \cos\left(\sqrt{\frac{(2+\sqrt{2})k}{m}}t + \phi_3\right) + A_2 \cos\left(\sqrt{\frac{(2-\sqrt{2})k}{m}}t + \phi_2\right) \right) \quad (25)$$

$$|2\rangle = \frac{1}{4} \left(-\sqrt{2}A_3 \cos\left(\sqrt{\frac{(2+\sqrt{2})k}{m}}t + \phi_3\right) + \sqrt{2}A_2 \cos\left(\sqrt{\frac{(2-\sqrt{2})k}{m}}t + \phi_2\right) \right) \quad (26)$$

$$|3\rangle = \frac{1}{4} \left(2A_1 \cos\left(\sqrt{\frac{2k}{m}}t + \phi_1\right) + A_3 \cos\left(\sqrt{\frac{(2+\sqrt{2})k}{m}}t + \phi_2\right) + A_2 \cos\left(\sqrt{\frac{(2-\sqrt{2})k}{m}}t + \phi_3\right) \right) \quad (27)$$

Given the 6 boundary conditions one can find the position of each mass at any given time. $\square$

Problem 2

We want to solve for the eigenvalues of the matrix below:

$$\Omega = \begin{pmatrix} -\frac{2k}{m} & \frac{k}{m} \\ \frac{k}{m} & -\frac{2k}{m} \end{pmatrix} \quad (28)$$

To do so we first calculate the following determinant:

$$\det(\Omega - \lambda I) \quad (29)$$

which is:

$$= \left(\frac{2k}{m} + \lambda\right)\left(\frac{2k}{m} + \lambda\right) - \frac{k^2}{m^2} \quad (30)$$

$$= \lambda^2 + \frac{4k}{m}\lambda + \frac{4k^2}{m^2} - \frac{k^2}{m^2} \quad (31)$$

$$= \lambda^2 + \frac{4k}{m}\lambda + \frac{3k^2}{m^2} \quad (32)$$

setting this to zero, and solving for lambda:

$$0 = \lambda^2 + \frac{4k}{m}\lambda + \frac{3k^2}{m^2} \quad (33)$$

$$\lambda = \begin{cases} -\frac{3k}{m} \\ -\frac{k}{m} \end{cases} \quad (34)$$

Now solving:

$$\Omega |I\rangle = \lambda_I |I\rangle \quad (35)$$

leads to:

$$\Omega \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -2\frac{ak}{m} + \frac{bk}{m} \\ \frac{ak}{m} - \frac{2bk}{m} \end{pmatrix} \quad (36)$$

$$\rightarrow \begin{pmatrix} -2\frac{ak}{m} + \frac{bk}{m} \\ \frac{ak}{m} - \frac{2bk}{m} \end{pmatrix} = -\frac{3k}{m} \begin{pmatrix} a \\ b \end{pmatrix} \quad (37)$$

and similarly for λ_{II} and $|II\rangle$:

$$\Omega \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -2\frac{ak}{m} + \frac{bk}{m} \\ \frac{ak}{m} - \frac{2bk}{m} \end{pmatrix} \quad (38)$$

$$\rightarrow \begin{pmatrix} -2\frac{ak}{m} + \frac{bk}{m} \\ \frac{ak}{m} - \frac{2bk}{m} \end{pmatrix} = -\frac{k}{m} \begin{pmatrix} a \\ b \end{pmatrix} \quad (39)$$

solving the first system of equations leads to:

$$|I\rangle = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \quad (40)$$

and the second one:

$$|II\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (41)$$

Which if normalized would become:

$$|I\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix} \quad (42)$$

$$|II\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (43)$$

$$\square \quad (44)$$

Problem 3

One can write the two eigenstates as:

$$|I\rangle = \frac{1}{\sqrt{2}} (-|1\rangle + |2\rangle) \quad (45)$$

$$|II\rangle = \frac{1}{\sqrt{2}} (|1\rangle + |2\rangle) \quad (46)$$

where:

$$|1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (47)$$

Therefore, for the coupled spring problem since:

$$|I\rangle = A_I \cos\left(\sqrt{\frac{3k}{m}}t\right) \quad (48)$$

$$|II\rangle = A_{II} \cos\left(\sqrt{\frac{k}{m}}t\right) \quad (49)$$

We solve 45 and 46 for $|1\rangle, |2\rangle$ we get:

$$|1\rangle = \frac{1}{2} \left(-A_I \cos \left(\sqrt{\frac{3k}{m}} t \right) + A_{II} \cos \left(\sqrt{\frac{k}{m}} t \right) \right) \quad (50)$$

$$|2\rangle = \frac{1}{2} \left(A_I \cos \left(\sqrt{\frac{3k}{m}} t \right) + A_{II} \cos \left(\sqrt{\frac{k}{m}} t \right) \right) \quad (51)$$

$$\square \quad (52)$$

Problem 4

I couldn't see any "matrix" above.

Problem 5

We assume Ω is a Hermitian operator with $\|\Omega\| < 1$. Consider the partial sums

$$S_N = \sum_{n=0}^N \Omega^n. \quad (53)$$

Multiply by $(I - \Omega)$:

$$(I - \Omega)S_N = (I - \Omega)(I + \Omega + \Omega^2 + \dots + \Omega^N) \quad (54)$$

$$= I - \Omega^{N+1}. \quad (55)$$

Since $\|\Omega\| < 1$, we have

$$\|\Omega^{N+1}\| \leq \|\Omega\|^{N+1} \longrightarrow 0 \quad (N \rightarrow \infty). \quad (56)$$

Taking the limit of both sides,

$$(I - \Omega) \left(\sum_{n=0}^{\infty} \Omega^n \right) = I. \quad (57)$$

Hence the series converges and its sum is the inverse of $(I - \Omega)$:

$$\sum_{n=0}^{\infty} \Omega^n = (I - \Omega)^{-1}. \quad (58)$$

Thus the geometric series formula holds for Hermitian operators whose norm is strictly less than 1.

Problem 6

Recall that H is Hermitian if $H^\dagger = H$, and U is unitary if $U^\dagger U = I$. We need to show that $(e^{iH})^\dagger e^{iH} = I$. Therefore, first we calculate the adjoint of e^{iH} :

$$(e^{iH})^\dagger = \left(\sum_{n=0}^{\infty} \frac{(iH)^n}{n!} \right)^\dagger \quad (59)$$

$$= \sum_{n=0}^{\infty} \frac{(iH)^n}{n!}^\dagger \quad (60)$$

$$= \sum_{n=0}^{\infty} \frac{(i^*)^n (H^\dagger)^n}{n!} \quad (61)$$

$$= \sum_{n=0}^{\infty} \frac{(-i)^n H^n}{n!} \quad (62)$$

$$= e^{-iH} \quad (63)$$

Now the product:

$$(e^{iH})^\dagger e^{iH} = e^{-iH} e^{iH} \tag{64}$$

$$= e^{-iH+iH} \tag{65}$$

$$= e^0 \tag{66}$$

$$= I \tag{67}$$

Therefore, e^{iH} is unitary. $\square$